

CASCADING STYLESHEET

Styling your HTML pages

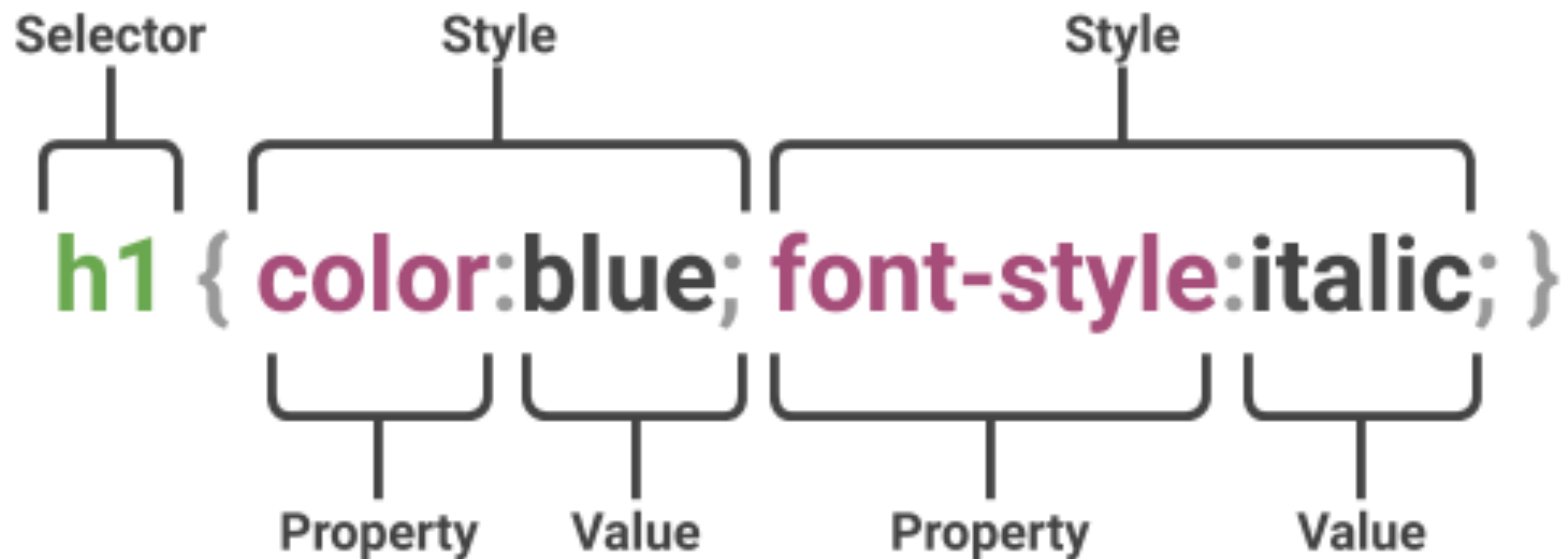
WHAT IS CSS?

- CSS stands for Cascading Style Sheet.
- CSS is used to design HTML tags.
- CSS is a widely used language on the web.
- is used in combination with HTML that customizes how HTML elements will appear. CSS can define styles and change the layout and design of a sheet.

```
<h2 style="text-align: center;">Centered text</h2>
```

```
<p style="color: blue; font-size: 18px;">Blue, 18-point text</p>
```

CSS SYNTAX



WHERE CAN YOU WRITE YOUR CSS ?

Inline CSS

```
<h2 style="text-align:
center;">Centered text</h2>

<p style="color: blue; font-size:
18px;">Blue, 18-point text</p>
```

Internal CSS

```
<head>
  <style>
    h1 {
      color: blue;
    }
  </style>
</head>
```

External CSS

```
<head>
  <link href="style.css"
type="text/css" rel="stylesheet">
</head>
```

The `<link>` element is used to link HTML documents to external resources like CSS files. It commonly uses:

- **href** attribute to specify the URL to the external resource
- **rel** attribute to specify the relationship of the linked document to the current document
- **type** attribute to define the type of content being linked

2 MAJORS DIFFICULTIES IN USING CSS

CSS Selector

How you select the HTML element to apply your style

1. CSS Element Selector
2. CSS Id Selector
3. CSS Class Selector
4. CSS Universal Selector
5. CSS Attribute Selector
6. CSS Group Selector

CSS Properties

How you choose the CSS properties effectively

- | | |
|--------------------|-----------------------|
| 1. CSS Background | 8. CSS Padding/Margin |
| 2. CSS Border | 9. CSS Position |
| 3. CSS Display | 10. CSS Visibility |
| 4. CSS Float | 11. CSS Width/Height |
| 5. CSS Font | 12. CSS Align |
| 6. CSS Line Height | 13. CSS Word wrap |
| 7. CSS Opacity | |

BASIC CSS SELECTOR: ELEMENT SELECTOR

The CSS **type or element selector** matches elements by node name. In other words, it selects all elements of the given type within a document.

```
span {  
  background-color: skyblue;  
}
```

HTML

```
<span>Here's a span with some text.</span>  
<p>Here's a p with some text.</p>  
<span>Here's a span with more text.</span>
```

Result

Here's a span with some text.

Here's a p with some text.

Here's a span with more text.

BASIC CSS SELECTOR: ID SELECTOR

The CSS **ID selector** matches an element based on the value of the element's [id](#) attribute. In order for the element to be selected, its id attribute must match exactly the value given in the selector.

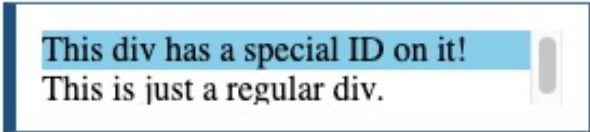
CSS

```
#identified {  
  background-color: skyblue;  
}
```

HTML

```
<div id="identified">This div has a special ID on it!</div>  
<div>This is just a regular div.</div>
```

Result



This div has a special ID on it!
This is just a regular div.

BASIC CSS SELECTOR: CLASS SELECTOR

The [CSS class selector](#) matches elements based on the contents of their [class](#) attribute.

CSS

```
.red {  
  color: #f33;  
}  
  
.yellow-bg {  
  background: #ffa;  
}  
  
.fancy {  
  font-weight: bold;  
  text-shadow: 4px 4px 3px #77f;  
}
```

Result

This paragraph has red text.

This paragraph has red text and a yellow background.

This paragraph has red text and "fancy" styling.

This is just a regular paragraph.

HTML

```
<p class="red">This paragraph has red text.</p>  
<p class="red yellow-bg">This paragraph has red text and a yellow backgr  
<p class="red fancy">This paragraph has red text and "fancy" styling.</p>  
<p>This is just a regular paragraph.</p>
```


BASIC CSS SELECTOR: ATTRIBUTE SELECTOR

The CSS **attribute selector** matches elements based on the presence or value of a given attribute.

CSS

```
a {  
  color: blue;  
}  
  
/* Internal links, beginning with "#" */  
a[href^="#"] {  
  background-color: gold;  
}  
  
/* Links with "example" anywhere in the URL */  
a[href*="example"] {  
  background-color: silver;  
}  
  
/* Links with "insensitive" anywhere in the URL,  
   regardless of capitalization */  
a[href*="insensitive" i] {  
  color: cyan;  
}  
  
/* Links with "cAsE" anywhere in the URL,  
   with matching capitalization */  
a[href*="cAsE" s] {  
  color: pink;  
}
```

```
/* Links that end in ".org" */  
a[href$=".org"] {  
  color: red;  
}  
  
/* Links that start with "https" and end in ".org" */  
a[href^="https"][href$=".org"] {  
  color: green;  
}
```

Result

- [Internal link](#)
- [Example link](#)
- [Insensitive internal link](#)
- [Example org link](#)
- [Example https org link](#)

HTML

```
<ul>  
  <li><a href="#internal">Internal link</a></li>  
  <li><a href="http://example.com">Example link</a></li>  
  <li><a href="#InSensitive">Insensitive internal link</a></li>  
  <li><a href="http://example.org">Example org link</a></li>  
  <li><a href="https://example.org">Example https org link</a></li>  
</ul>
```

BASIC CSS SELECTOR: UNIVERSAL SELECTOR

The CSS **universal selector** (*) matches elements of any type.

CSS

```
* [lang^=en] {
  color: green;
}

*.warning {
  color: red;
}

*#maincontent {
  border: 1px solid blue;
}

.floating {
  float: left;
}

/* automatically clear the next sibling after a floating element */
.floating + * {
  clear: left;
}
```

HTML

```
<p class="warning">
  <span lang="en-us">A green span</span> in a red paragraph.
</p>
<p id="maincontent" lang="en-gb">
  <span class="warning">A red span</span> in a green paragraph.
</p>
```

Result

A green span in a red paragraph.

A red span in a green paragraph.

GROUP CSS SELECTOR

The , selector is a grouping method that selects all the matching nodes. **Syntax:** A, B **Example:** div, span will match both [](#) and [<div>](#) elements.

```
h1, h2, h3, h4, h5, h6 { font-family: helvetica; }
```

COMBINATOR CSS SELECTOR

You can apply more condition into your selection. This will help in complicate scenario where you only want to apply your CSS to a section of your pages only. There are different combinators:

Descendant combinator: The (space) combinator selects nodes that are descendants of the first element. **Syntax:** A B

Child combinator: The > combinator selects nodes that are direct children of the first element. **Syntax:** A > B

General sibling combinator: The ~ combinator selects siblings. This means that the second element follows the first (though not necessarily immediately), and both share the same parent. **Syntax:** A ~ B

Adjacent sibling combinator: The + combinator matches the second element only if it *immediately* follows the first element. **Syntax:** A + B

Column combinator: The || combinator selects nodes which belong to a column. **Syntax:** A || B

PSEUDO-CLASS CSS SELECTOR

A [CSS](#) **pseudo-class** is a keyword added to a selector that specifies a special state of the selected element(s). For example, [:hover](#) can be used to change a button's color when the user's pointer hovers over it.

```
/* Any button over which the user's pointer is hovering */
button:hover {
  color: blue;
}
```

PSEUDO-ELEMENTS CSS SELECTOR

A CSS **pseudo-element** is a keyword added to a selector that lets you style a specific part of the selected element(s). For example, [::first-line](#) can be used to change the font of the first line of a paragraph.

Syntax

```
selector::pseudo-element {  
  property: value;  
}
```

```
/* The first line of every <p> element. */  
p::first-line {  
  color: blue;  
  text-transform: uppercase;  
}
```

More detail: <https://developer.mozilla.org/en-US/docs/Web/CSS/Pseudo-elements>

CSS PROPERTIES: THE BASIC PROPERTIES

A *CSS property* styles an aspect of an HTML element.

A CSS property declaration consists of a property name and a property value. The property name comes first, then a colon, and then the value. Here is the general pattern a CSS property declaration follows:

```
property-name : property-value
```


MORE ADVANCED CSS PROPERTIES ARE
COMING LATER